

The Fisher–KPP Traveling-Wave Atlas for Every Real Speed: Threshold Fronts, Oscillatory Obstructions, and the Route-A Stop

Route-A structural certificate HCS-C202

Abstract

For all diffusion and growth coefficients $D, r > 0$, we classify every real speed in the Fisher–KPP traveling-wave equation. A dimensionless phase-plane triangle proves existence and translation uniqueness of positive monotone fronts exactly when $c \geq 2\sqrt{Dr}$.

Keywords: Fisher–KPP equation; traveling front; minimal speed; phase plane; heteroclinic orbit; reaction–diffusion.

中文摘要

对任意扩散系数与增长率 $D, r > 0$ ，本文统一分类 Fisher–KPP 行波方程的全部实速度。相平面不变三角形证明正单调波前恰在 $c \geq 2\sqrt{Dr}$ 时存在，且除平移外唯一。负速度由空间反射得到；亚临界速度的焦点尾部必然变号， $c = 0$ 虽有哈密顿周期卵形，却不存在所需异宿轨道。三种临界尾部与精确特殊解共同锁定全部符号和尺度；有限计算只作回归，不替代连续存在性证明。该模型没有内生素数载体、周期轨道层或目标谱结构，故严格否决路线 A。

关键词：Fisher–KPP 方程；行波；最小速度；相图；异宿轨道。

1 Frozen equation and exact scaling

Fix $D, r > 0$ in physical time and consider

$$u_t = Du_{xx} + ru(1 - u). \quad (1)$$

Fisher introduced this logistic diffusion model and its wave argument [1]; Kolmogorov, Petrovskii and Piskunov established the classical minimal-speed front theory [2]. With $z = x - ct$, $u(x, t) = U(z)$, (1) becomes

$$DU_{zz} + cU_z + rU(1 - U) = 0. \quad (2)$$

Set

$$\xi = \sqrt{r/D} z, \quad s = c/\sqrt{Dr}, \quad V = U_\xi.$$

The complete parameter family is conjugate, with time orientation retained, to

$$U_\xi = V, \quad V_\xi = -sV - U(1 - U). \quad (3)$$

The equilibria are $(1, 0)$ and $(0, 0)$. The first is always a saddle; the zero-state characteristic polynomial is

$$\lambda^2 + s\lambda + 1. \quad (4)$$

2 The positive-speed front theorem

Assume $s \geq 2$ and define the slow positive slope

$$q = \frac{s - \sqrt{s^2 - 4}}{2}, \quad q^2 - sq + 1 = 0.$$

Consider the compact triangle

$$\mathcal{T}_q = \{(U, V) : 0 \leq U \leq 1, -qU \leq V \leq 0\}. \quad (5)$$

On $V = 0$, $V_\xi = -U(1 - U) < 0$, while on $U = 1$ one has $U_\xi = V \leq 0$. On the slanted boundary put $G = V + qU$; substitution of $V = -qU$ gives the decisive identity

$$G_\xi = U\{q(s - q) - 1 + U\} = U^2 > 0. \quad (6)$$

The decreasing branch of the one-dimensional unstable manifold of $(1, 0)$ enters $\mathcal{T}_q$. Since the divergence of (3) is the negative constant $-s$, a nonconstant periodic omega limit is impossible. The energy identity below also prevents return to the saddle; hence the branch converges to $(0, 0)$. It is a profile with

$$0 < U < 1, \quad U_\xi < 0, \quad U(-\infty) = 1, \quad U(+\infty) = 0. \quad (7)$$

Conversely, every orbit with (7) must leave the saddle along this same unstable branch. Reparametrizing one orbit changes only the origin of ξ . Thus for every $c \geq 2\sqrt{Dr}$ the front exists and is unique modulo translation.

Revision focus. Round 0 freezes all conventions and closes the critical/supercritical trapping and translation-uniqueness theorem.

Declarations

Data and code availability. Package-local deterministic code and synthetic exact evidence accompany this manuscript.

Ethics. No human, animal, clinical, personal or sensitive data are used; approval is not applicable. Historical sources are cited for mathematical provenance, not ideological endorsement.

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